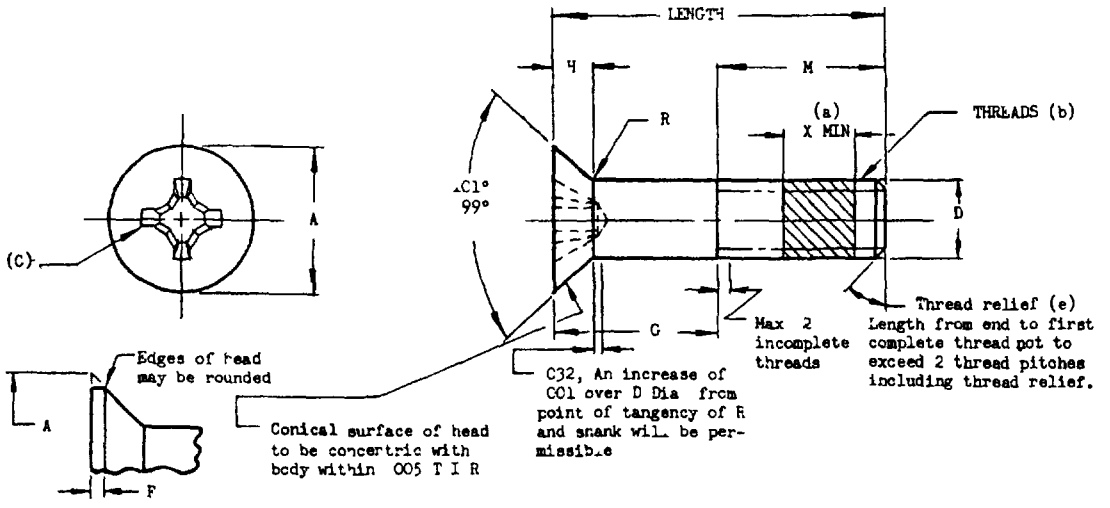
FED SUP CLASS
5306



DASH NO DIA	(b) THREADS	D DIA MAX	D DIA MIN	A DIA MAX SHARP	A DIA MIN SHARP	A DIA ABSOLUTE MIN WITH MAX F	F MAX	(d) H	M REF	R RAD MAX	R RAD MIN	(f) TENSILE STRENGTH LBS MIN	DOUBLE SHEAR STRENGTH LBS MIN	DRIVER SIZE
04	4-40UNC-3A	.1120	.0925	.225	.207	.200	.010	.044	.250	.015	.005	760	1,480	1
06	6-32UNC-3A	.1380	.1141	.279	.257	.249	.014	.055	.312			1,120	2,240	
08	8-32UNC-3A	.1640	.1610	.332	.319	.283	.015	.068	.437			1,740	3,170	2
3	10-32UNF-3A	.1890	.1860	.385	.371	.327	.016	.080	.469			2,420	4,250	
4	1/4-28UNF-3A	.2490	.2460	.507	.491	.447	.018	.106	.506			4,420	7,360	3
5	5/16-24UNF-3A	.3115	.3085	.635	.617	.569	.020	.133	.531			7,100	11,500	
6	3/8-24UNF-3A	.3740	.3710	.762	.742	.686	.023	.159	.641			10,790	16,560	4
7	7/16-20UNF-3A	.4365	.4325	.890	.868	.805	.026	.186	.656	.020	.010	14,600	22,500	
8	1/2-20UNF-3A	.4990	.4950	1.017	.992	.919	.030	.213	.781			19,810	29,400	
9	9/16-18UNF-3A	.5615	.5575	1.145	1.118	1.036	.034	.240	.906			25,060	37,400	

(a) Location of effective locking area X shall be in accordance with MS15981
(b) Threads shall be formed by any single rolling process and in accordance with Specification MIL-S-7742
(c) Dimensioning and gaging of recess design shall be in accordance with MS9006
(d) Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement
(e) Chamfer 45° ±5° (See MIL-STD-8 for clarification of angular tolerance) Chamfer to extend to root of thread
(f) Based on 125,000 PSI ultimate tensile strength, and stress area = 3.1416 (E/2 - 3H/.16)^2 See Handbook H-28 (1957) Part 1

MATERIAL Alloy Steel - See Procurement Specification
HARDNESS Rockwell C26-C32
PLATING Cadmium Plate, QQ-P-416, Type II, Class 3
SURFACE ROUGHNESS The surface of the shank, fillet and bearing area shall not exceed .25 micro-inches See MIL-STD-10
LOCKING ELEMENT In accordance with Specification MIL-F-18240 Locking element area, with locking element incorporated must pass thru a ring gage equal to maximum major thread diameter + .003" freely or with finger pressure
DESIGN AND USAGE LIMITATIONS See MS15981

EXAMPLE OF PART NUMBER MS21091-4008 = 1/4-28 bolt, 500 grip

DIMENSIONS In inches

FOR DESIGN FEATURE PURPOSES THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID

Review activities
Navy - WP
USAF - 11

This standard has been approved by the Bureau (Department of the Navy) and the Department of the USAF and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable

APPROVED 2 Nov 65 REVISED

P A Navy - WP Other Cust USAF - 11	TITLE BOLT, SELF-LOCKING, 250° F, STEEL, 75 KSI Fsu, 125 KSI Ftu, 100° FLUSH HEAD, CROSS RECESSED	MILITARY STANDARD MS21091(ASG)
PROCUREMENT SPECIFICATION MIL-S-7839 & MIL-F-18240	SUPERSEDES	SHEET 1 OF 2

DD FORM 672-1 PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

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LENGTH DASH NO.(a)	GRIP G ±.010	LENGTH +.031 -.015									
		4-40	6-32	8-32	10-32	1/4-28	5/16-24	3/8-24	7/16-20	1/2-20	9/16-18
002	.125	.375	.437	.562							
003	.187	.437	.500	.625	.593						
004	.250	.500	.562	.687	.656	.719					
006	.375	.625	.687	.812	.781	.844	.906	1.016			
008	.500	.750	.812	.937	.906	.969	1.031	1.141	1.156	1.281	
010	.625	.875	.937	1.062	1.031	1.094	1.156	1.266	1.281	1.406	1.531
012	.750	1.000	1.062	1.187	1.156	1.219	1.281	1.391	1.406	1.531	1.656
014	.875	1.125	1.187	1.312	1.281	1.344	1.406	1.516	1.531	1.656	1.781
016	1.000	1.250	1.312	1.437	1.406	1.469	1.531	1.641	1.656	1.781	1.906
018	1.125	1.375	1.437	1.562	1.531	1.594	1.656	1.766	1.781	1.906	2.031
020	1.250	1.500	1.562	1.687	1.656	1.719	1.781	1.891	1.906	2.031	2.156
022	1.375	1.625	1.687	1.812	1.781	1.844	1.906	2.016	2.031	2.156	2.281
024	1.500	1.750	1.812	1.937	1.906	1.969	2.031	2.141	2.156	2.281	2.406
026	1.625	1.875	1.937	2.062	2.031	2.094	2.156	2.266	2.281	2.406	2.531
028	1.750	2.000	2.062	2.187	2.156	2.219	2.281	2.391	2.406	2.531	2.656
030	1.875		2.187	2.312	2.281	2.344	2.406	2.516	2.531	2.656	2.781
032	2.000		2.312	2.437	2.406	2.469	2.531	2.641	2.656	2.781	2.906
034	2.125		2.437	2.562	2.531	2.594	2.656	2.766	2.781	2.906	3.031
036	2.250		2.562	2.687	2.656	2.719	2.781	2.891	2.906	3.031	3.156
038	2.375		2.687	2.812	2.781	2.844	2.906	3.016	3.031	3.156	3.281
040	2.500		2.812	2.937	2.906	2.969	3.031	3.141	3.156	3.281	3.406
042	2.625				3.031	3.094	3.156	3.266	3.281	3.406	3.531
044	2.750				3.156	3.219	3.281	3.391	3.406	3.531	3.656
046	2.875				3.281	3.344	3.406	3.516	3.531	3.656	3.781
048	3.000				3.406	3.469	3.531	3.641	3.656	3.781	3.906
050	3.125				3.531	3.594	3.656	3.766	3.781	3.906	4.031
052	3.250									4.031	4.156
054	3.375									4.156	4.281
056	3.500									4.281	4.406
058	3.625									4.406	4.531
060	3.750									4.531	4.656
062	3.875									4.656	4.781
064	4.000									4.781	4.906
066	4.125									4.906	5.031

(a) Unassigned dash numbers shall not be used except those dash numbers included by the following coding note: Dash numbers indicate preferred grip lengths in .125 increments for lengths longer than 2.50 inches. Intermediate grip lengths between those lengths listed may be specified in .062 increments by intermediate dash numbers. Length and grip of a bolt with an odd number for the dash number will be .062 greater than shown for the bolt with the preceding even dash number.

APPROVED 2 Nov 65 REVISED

P.A. Navy - WP Other Cust USAF - 11	TITLE BOLT, SELF-LOCKING, 250° F, STEEL 75 KSI Fsu, 125 KSI Ftu, 100° FLUSH HEAD, CROSS RECESSED	MILITARY STANDARD MS21091 (ASG)
PROCUREMENT SPECIFICATION MIL-S-7839 & MIL-F-18240	SUPERSEDES	SHEET 2 OF

DD FORM 1 SEP 67 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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